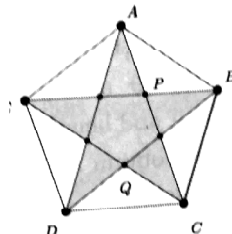
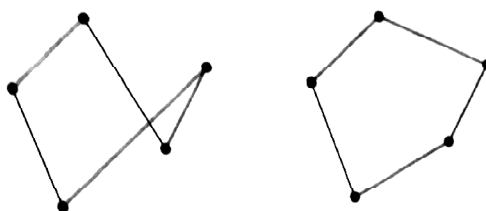


PART - A

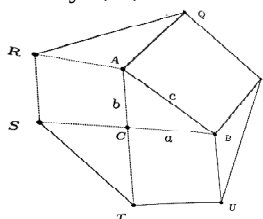
1. P is a point on AL, the altitude of the triangle ABC through A. If $\angle PBA = 20^\circ$, $\angle PBC = 40^\circ$ and $\angle PCB = 30^\circ$, then $\angle PCA$ equals
(A) 20° (B) 10° (C) 15° (D) 18°
2. ABCDE is a regular pentagon. The area of the star shaped region ACEBDA is 1 square cm. AC and BE meet at P and BD and CE meet at Q as shown in the figure. The area of APQD in square cms is



- (A) $\frac{1}{4}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{3}{4}$
3. The number of integer pairs (x, y) which satisfy the equation $x^3 = y^3 + 2y^2 + 1$ is
(A) 0 (B) 1 (C) 2 (D) 3
4. Let $f(x) = x^4 + x^3 + x^2 + x + 1$. The remainder when $f(x^5)$ is divided by $f(x)$ is
(A) 5 (B) 6 (C) 7 (D) 4
5. The number of positive integers a, b, c such that $a^2 + b^2 + c^2 = a^2b^2$ is
(A) 1 (B) 2 (C) 5 (D) none of these
6. The number of triples (x, y, z) of real numbers such that $3x^2 + y^2 + z^2 = 2x(y + z)$ is
(A) 1 (B) 2 (C) 3 (D) 5
7. The number of real roots of the equation $x^4 - 4x = 1$ is
(A) 1 (B) 2 (C) 0 (D) 4
8. Given a set of r points in the plane so that no three are collinear, by a closed polygon we mean the polygon obtained by connecting them by r line segments as shown in the examples below in Figure 2 (here $r = 5$). There are 10 points on a plane no three of which are collinear. The number of 5 sided closed polygons whose vertices are among these 10 points is



- (A) 6048 (B) 1507 (C) 3024 (D) 10000
9. In the adjacent figure ABC is a right angled triangle. Squares are described externally on its sides and the outer vertices of these squares are joined as shown. If the lengths of the sides AB, BC, CA are respectively c, a, b the area of the hexagon PQRSTU is



- (A) $2(a^2 + ab + b^2)$ (B) $a^2 + ab + b^2$
- (C) $3(a^2 + ab + b^2)$ (D) $\frac{1}{2}(a^2 + ab + b^2)$

10. ABCD is a square. From B, D lines are drawn to meet at P inside the square such that $\angle ADP = 25^\circ$ and $\angle ABP = 20^\circ$. Then $\angle BPC$ is
 (A) 70° (B) 80° (C) 60° (D) 50°
11. If p, q are positive odd integers such that $(1 + 3 + 5 + \dots + p) + (1 + 3 + 5 + \dots + q) = 1 + 3 + \dots + 19$ then $p + q$ is
 (A) a prime number (B) divisible by 13
 (C) odd number (D) none of these
12. The number $2^{20} - 1$ is divisible by
 (A) 11 and 41 (B) 11 and 21 (C) 41 and 61 (D) 11 and 61
13. Five points O, A, B, C, D are taken in order on a straight line such that $OA = a$, $OB = b$, $OC = c$ and $OD = d$. P is a point on the line between B and C. If $AP : PD = BP : BC$, then OP is
 (A) $\frac{ac - bd}{a - b + c - d}$ (B) $\frac{ac + bd}{a - b + c - d}$ (C) $\frac{ad - bc}{a - b + c - d}$ (D) none of these
14. The side AB of an equilateral triangle AB is produced to D such that $BD = 2AB$. The point F is the foot of the perpendicular from D on CB produced. $\angle FAC =$
 (A) 70° (B) 75° (C) 80° (D) 90°
15. For the simultaneous equations $x^2 + 2xy + y^2 - x - y = 6$, $x - 2y = 3$
 (A) there is a solution (x, y) such that both x, y are irrational
 (B) there are two sets of solutions (x, y) such that x, y are integers
 (C) sum of all solutions is 1
 (D) product of all solutions is $\frac{5}{3}$

PART - B

16. The number of right angled triangles with integer side lengths and such that the product of the lengths of the legs (non-hypotenuse sides) equals three times the perimeter of the triangle is _____.
17. $\alpha, \beta, \gamma, \delta$ are the roots of the equation $x^4 - ax^3 + ax^2 + bx + c = 0$, where a, b, c are real numbers. The smallest possible value of $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$ is _____.
18. Two circles with centers P and Q and radii 3 and 4 respectively touch each other externally. AB, CD are direct common tangents touching the smaller circle at A, C and the bigger circle at B, D. The area of the concave hexagon APCDQB is _____.
19. If a, b are the lengths of unequal diagonals of a regular heptagon (regular polygon with 7 sides) with side c, then $\frac{1}{a} + \frac{1}{b}$ in terms of c is _____.
20. The number of real roots of the equation $\frac{\log_{10}(\sqrt{x+1}+1)}{\log_{10}\sqrt[3]{x-40}} = 3$ is _____.
21. x, y, z are non zero real numbers such that $x^2 + y^2 + z^2 = 1$
 $x\left(\frac{1}{y} + \frac{1}{z}\right) + y\left(\frac{1}{z} + \frac{1}{x}\right) + z\left(\frac{1}{x} + \frac{1}{y}\right) + 3 = 0$ The number of possible values of $x + y + z$ is _____.
22. The minimum value of integer n such that among any n integers we can always find three integers whose sum is divisible by 3 is _____.
23. The number of integers n for which $n^4 - 51n^2 + 50$ is negative is _____.
24. In a triangle ABC, the lengths of the sides are consecutive integers and the median drawn from A is perpendicular to the bisector of angle B. The largest side of the triangle has length _____.
25. a, b, c, d, e are real numbers such that

$$a + 4b + 9c + 16d + 25e = 1$$

$$4a + 9b + 16c + 25d + 36e = 8$$

$$9a + 16b + 25c + 36d + 49e = 23$$
 The value of $a + b + c + d + e$ is _____.

26. In a triangle ABC, the altitude, angle bisector and the median from C divide the angle C into four equal angles. The measure of the least angle of the triangle is _____.
27. AB is a chord of a circle with center O. AB is produced to C such that $BC = OA$. CO is produced to E. The value of $\frac{\angle AOE}{\angle ACE}$ is _____.
28. The number of two digit numbers that are less than the sum of the squares of their digits by 11 and exceed twice the product of their digits by 5 is _____.
29. ABD is a circle whose centre is C. The circle circumscribing ABC cuts DA or DA produced at E. Then the triangle BDE is a _____ triangle.
30. The number of 4-digit numbers N such that
- no digit of N is 9
 - N is the square of an integer
 - when each digit of N is increased by 1, the resulting number is also square of an integer

AMTI (NMTC) - 2017
GAUSS CONTEST - INTER LEVEL
(Standard - XI & XII)

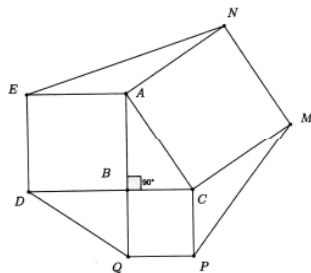
Note :

1. Fill in the response sheet with your Name, Class and the institution through which you appear in the specified places.
2. Diagrams are only visual aids; they are NOT drawn to scale.
3. You are free to do rough work on separate sheets.
4. Duration of the test : 2 pm to 4 pm – 2 hours.

PART - A

Note :

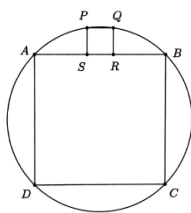
- ◆ Only one of the choices A, B, C, D is correct for each question. Shade the alphabet of your choice in the response sheet. If you have any doubt in the method of answering, seek the guidance of the supervisor.
 - ◆ For each correct response you get 1 mark. For each incorrect response you lose $\frac{1}{2}$ mark.
1. The equations $5x^2 - 10x \cos\alpha + 7\cos\alpha + 6 = 0$ has two identical roots. If α is one of the angles of a parallelogram with sum of the lengths of two adjacent sides equal to 6, the maximum area of the parallelogram is
(A) $32/5$ (B) $34/5$ (C) $36/5$ (D) $18/5$
 2. If $f(x)$ is a polynomial of degree three with leading coefficient 1 such that $f(1) = 1$, $f(2) = 4$, $f(3) = 9$, then the value of $f(4)$ is
(A) 16 (B) 22 (C) 36 (D) None of these
 3. The three roots of the equation $3x^3 + px^2 + qx - 4$ are the side length, inradius and the circumradius of an equilateral triangle. Then the value of $2p + q$ is
(A) -10 (B) 10 (C) -12 (D) 12
 4. Given $a = -\sqrt{99} + \sqrt{999} + \sqrt{9999}$
 $b = \sqrt{99} - \sqrt{999} + \sqrt{9999}$
 $c = \sqrt{99} + \sqrt{999} - \sqrt{9999}$ the value of $\frac{a^4}{(a-b)(a-c)} + \frac{b^4}{(b-c)(b-a)} + \frac{c^4}{(c-a)(c-b)}$
(A) 22194 (B) $\sqrt{99} + \sqrt{999} + \sqrt{9999}$
(C) 22190 (D) $\sqrt{99 \times 999 \times 9999}$
 5. In the figure, ABC is a right angled triangle with $\angle B = 90^\circ$, AB = 8 cm and BC = 6 cms. Squares ANMC, AEDB, BQPC are described as shown, on the sides AC, AB, BC respectively. Among PM, NE and DQ.



- (A) Two are of integral length and the length of the other is irrational
- (B) Two are of irrational length and the length of the other is integral
- (C) All have irrational lengths
- (D) All have integral length



6. In the adjoining figure, ABCD and PQRS are squares. If the length of the side of the bigger square is a , the length of the side of the smaller square is



- (A) $\frac{a}{3}$ (B) $\frac{a}{4}$ (C) $\frac{a}{5}$ (D) $\frac{a}{6}$
7. Twelve people sit around a circular table. Each observes that his age (viewed as an integer) is the average of the ages of his left and right neighbours. Which of the following could be the sum of their ages ?
 (A) 224 (B) 226 (C) 227 (D) 228
8. A rectangular billiard table has vertices at $(0, 0)$, $(12, 0)$, $(0, 10)$, $(12, 10)$. There are pockets only in the four corners. A ball is hit from the corner $(0, 0)$ along the line $y = x$ and bounces off several walls before eventually entering a pocket. The number of walls that the ball bounces off before entering a pocket is
 (A) 7 (B) 8 (C) 9 (D) 11
9. In a quadrilateral ABCD, we have $AB = 8$, $BC = 5$, $CD = 17$ and $DA = 10$. The diagonals AC and BD meet at E. If $BE = ED = 1 : 2$, the area of the quadrilateral ABCD is
 (A) 70 (B) 60
 (C) 50 (D) Not uniquely determined
10. Let $S(n)$ denote the sum of the digits of the integer n when n is written in the usual decimal form. For example, $S(123) = 6$. If $S(n) = 1274$, what is a possible value of $S(n + 1)$?
 (A) 1239 (B) 1266 (C) 1275 (D) 1284
11. The polynomial $g(x) = x^3 + ax^2 + bx + c$ has three distinct roots and each root of $g(x) = 0$ is also a root of $f(x) = x^4 + x^3 + bx^2 + 100x + c$. What is $f(1)$?
 (A) -5005 (B) -6006 (C) -7007 (D) -8008
12. The number $N = 1234567891011....41424344$ is the 79 digit number obtained by writing the numbers from 1 to 44 in order. What is the remainder when N is divided by 45 ?
 (A) 1 (B) 9 (C) 18 (D) 27
13. Define the sequence F_n recursively as follows : $F_0 = 1$, $F_1 = 1$ and for $n \geq 2$, F_n is the remainder of $F_{n-1} + F_{n-2}$ when divided by 3. What is the value of $\sum_{k=2017}^{2024} F_k$?
 (A) 6 (B) 7 (C) 8 (D) 9
14. P, Q, R, S have integer coordinates and are distinct points on the circle $x^2 + y^2 = 25$. The distance PQ and RS are irrational numbers. What is the largest possible value of PQ/RS ?
 (A) 3 (B) $5\sqrt{2}$ (C) 7 (D) $3\sqrt{5}$
15. In how many ways can 1000 be written as a sum of 2s and 3s, ignoring order ($500 \times 2 + 0 \times 3$ and $50 \times 2 + 300 \times 3$ are two of the ways) ?
 (A) 500 (B) 499 (C) 167 (D) 166

PART - B

Note :

- ♦ Write the correct answer in the space provided in the response sheet.
 - ♦ For each correct response you get 1 mark. For each incorrect response you lose $\frac{1}{4}$ mark.
16. The six digit number 789ABC consists of six distinct digits and is divisible by 7, 8 and 9. The three digit number ABC is _____.



17. m, n are relatively prime positive integers such that $\frac{m}{n} = \frac{2(\sqrt{2} + \sqrt{10})}{5\sqrt{3} + \sqrt{5}}$, then $m + n$ equals ____.
18. ABC is a triangle with $AB = 17$ units. F is the mid point of AB and $CF = 8$ units. The maximum possible area of the triangle ABC is ____.
19. Given that $a + b + c = 5$ and $1 \leq a, b, c \leq 2$ the minimum value of $\frac{1}{a+b} + \frac{1}{b+c}$ is ____.
20. The number of ways we can insert $+$'s between the digits of 111111111111111111 (eighteen 1's) so that the result will be a multiple of 90 is ____.
21. The largest positive integer less than 2017 that has exactly three proper factors (a proper factor is a factor other than the number of itself; for example, 11 has only one proper factor) is ____.
22. Consider the sequence 1, 3, 4, 7, 11, 18, 29, in which each term from the third term onwards is the sum of the two previous terms. Of the first 100 terms of this sequence the number of terms that are multiples of 5 is ____.
23. The non negative, distinct integers a, b, c, d, e form an arithmetic progression. If the sum of the numbers is 440, the maximum possible value of e is ____.
24. Let f be defined for all positive integers as follows : $f(n) = \begin{cases} n^2 + 1 & \text{if } n \text{ is odd} \\ \frac{n}{2} & \text{if } n \text{ is even} \end{cases}$ The number of integers n such that $1 \leq n \leq 100$ for which $f(f(...f(n))) = 1$ (where f is applied some number of times) is ____.
25. M and N are three digit numbers with no digit equal to zero. If we rearrange the digits of M all the numbers obtained are less than M . N has the same digits as M but in a different order. If we rearrange the digits of N , the largest number we obtain is M . If $M + N = 1233$, M equals ____.
26. $ABCDEF$ is a hexagon inscribed in a circle of radius R . If $AB = CD = EF = 2$ and $BC = DE = FA = 10$, the radius R is ____.
27. a, b, c are real numbers satisfying the following equations : $\log_2(abc - 3 + \log_5 a) = 5$
 $\log_3(abc - 3 + \log_5 b) = 4$
 $\log_4(abc - 3 + \log_5 c) = 4$
The value of $|\log_5 a| + |\log_5 b| + |\log_5 c|$ is ____.
28. In a triangle ABC , I is the incenter. The internal angle bisector of $\angle C$ meets AB at F and the circum circle of triangle ABC at Z . If $FI = 2$, $ZF = 3$ and $IC = m/n$ where m, n are relatively prime positive integers, the value of $m + n$ is ____.
29. The largest integer such that $n^3 + 4n^2 - 15n - 18$ is a perfect cube is ____.
30. The grid below shows a network of roads and pond (there are 8 horizontal lines and 8 vertical lines in the figure). You can move only horizontally or vertically from one grid point to an adjacent grid point. You do not know swimming and hence need to avoid going through the triangular pond at the top left corner of the grid. The number of shortest paths between P and Q is ____.

